

Module 3: Computing (Servers)

At the end of this module, you will be able to:

- **Describe** Amazon Elastic Compute Cloud (Amazon EC2) virtualization.
- **Differentiate** between the instance types and storage options available for EC2 instances.
- **Understand** the networking components that must be specified when you launch an EC2 instance.
- **Explain** Amazon EC2 user data and metadata.
- **Differentiate** the lifecycle states for an EC2 instance.
- **Explain** the shared responsibility model.
- **Create** Amazon EC2 instances.

Topics:

- Computing on AWS
- Managing your AWS instances
- Securing your AWS instances
- Pricing for Amazon EC2 instances

Knowledge check

Demo:

- Launch an Amazon EC2 instance, view system log, get instance screenshot

Lab:

- Lab 2: Creating Amazon EC2 instances

Exercises / Project:

- Activity 3: Troubleshoot creating a LAMP instance via AWS CLI
- Project time: Troubleshooting knowledge base

Troubleshooting knowledge Base – Module 3 focus

- Have your copy of the troubleshooting knowledge base template open for editing
- **Create new entries** as you encounter troubleshooting situations
- **Categories of focus** for this module:
 - Compute
 - Foundational IT

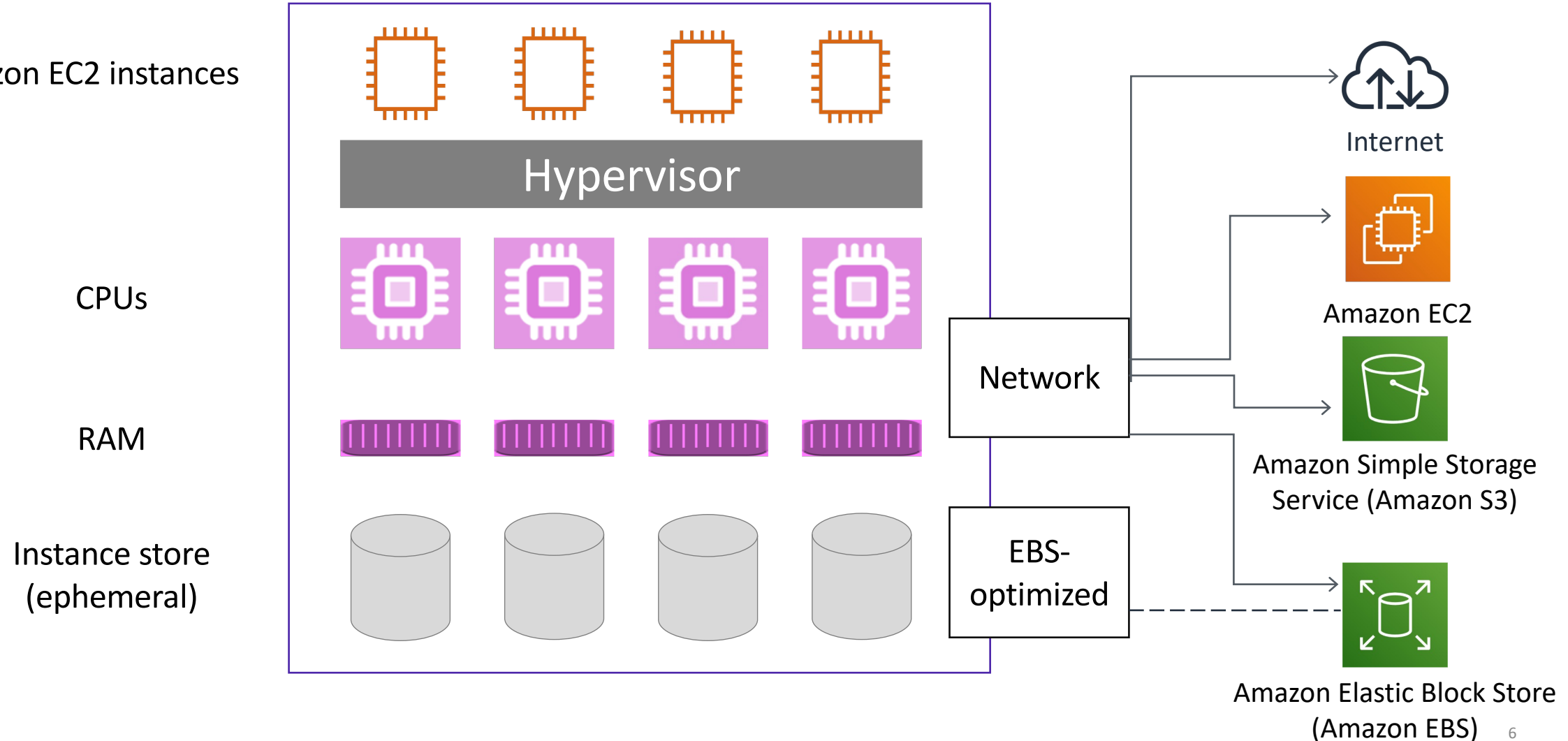
Module 3: Computing (Servers)

Computing on AWS



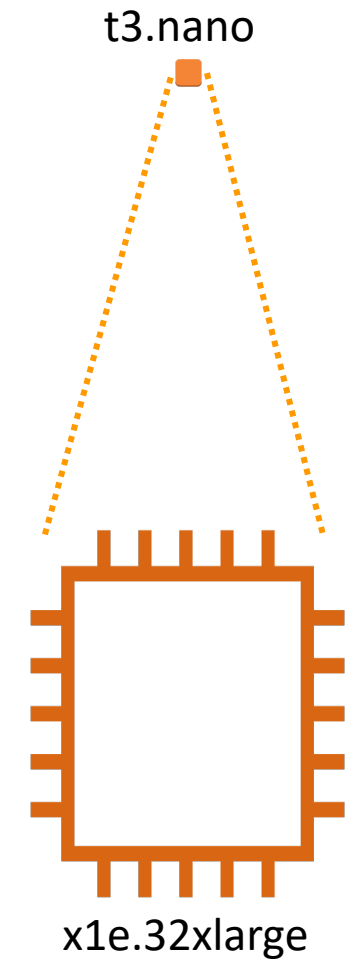
Amazon EC2 virtualization

Amazon EC2 instances

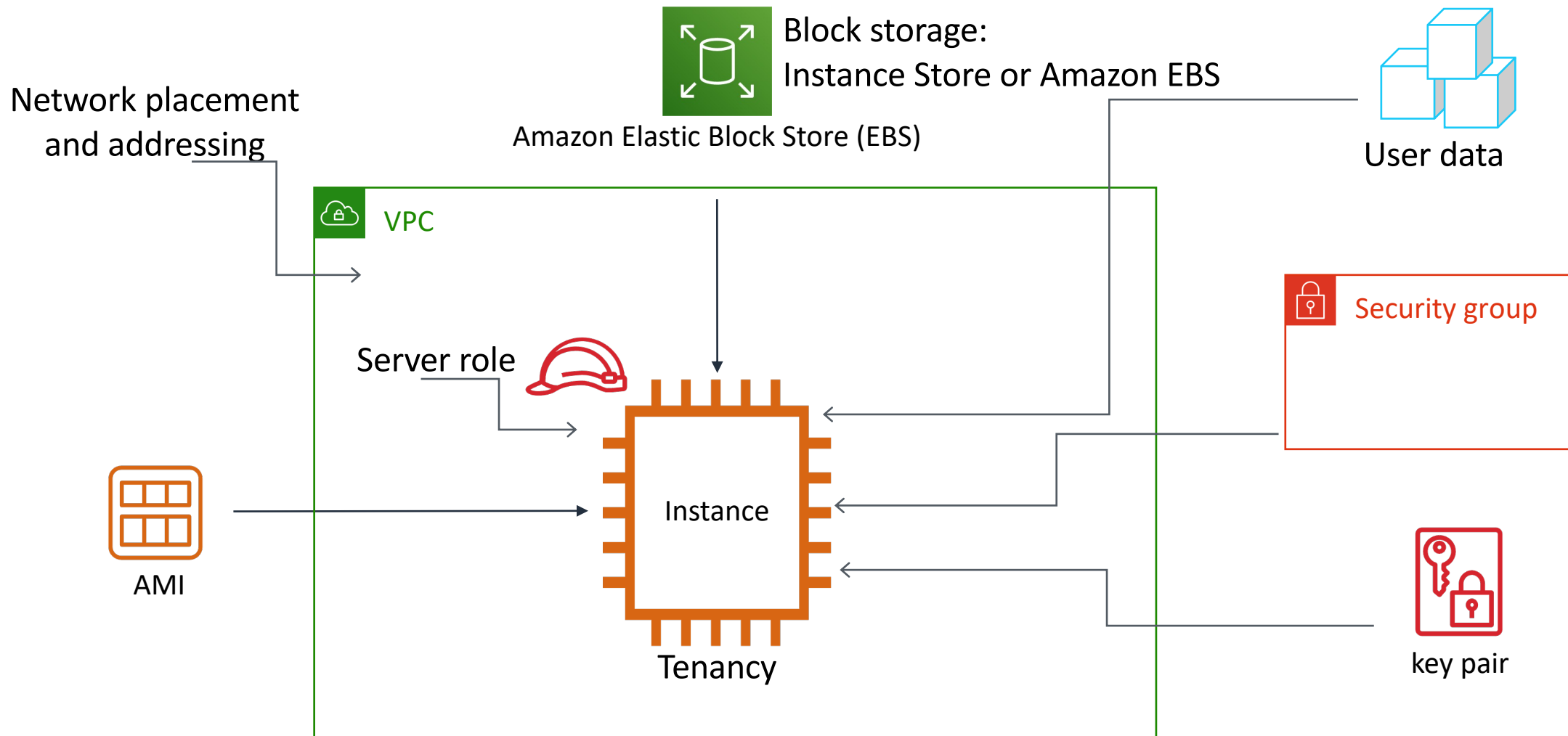


Instance types

Variations		
CPU type Storage type	CPU or core count Storage amount	Memory amount Network performance
Categories		
General purpose Accelerated computing	Compute optimized Storage optimized	Memory optimized
Families		
Collections of types within a category		
Types		
Unique selections of virtual hardware capability		



Provisioning an instance



**AMIs consist
of:**

AMI choices:

**Template for
instance
volumes**

**Launch
permissions**

**Block device
mappings**

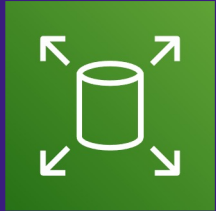
**AWS
Provided**

User Built

Community

Marketplace

Block storage for EC2 instances



Amazon Elastic Block Store (Amazon EBS)

Solid state drives (SSD):

Hard disk drives (HDD):

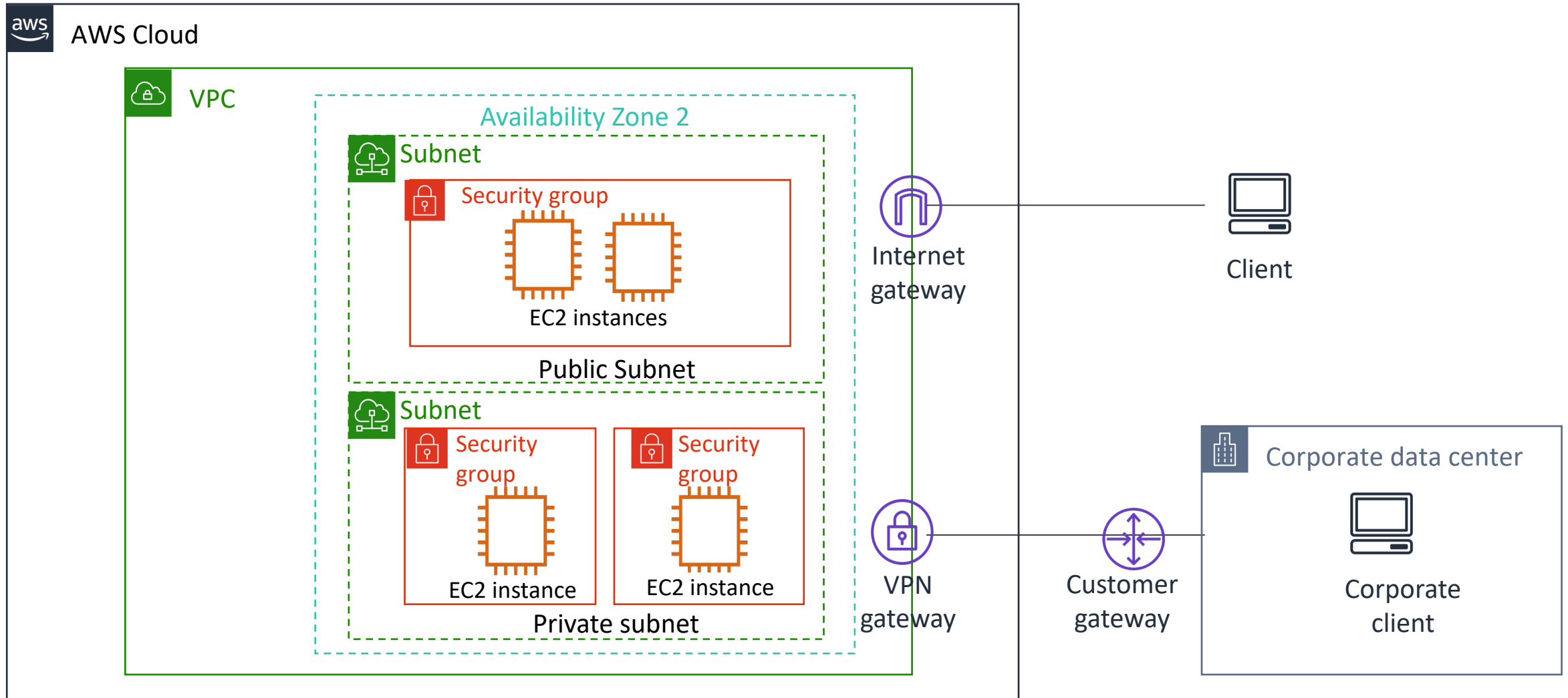
Provisioned IOPS
SSD (io1) volumes

General purpose
SSD (gp2) volumes

Throughput
optimized HDD
(st1) volumes

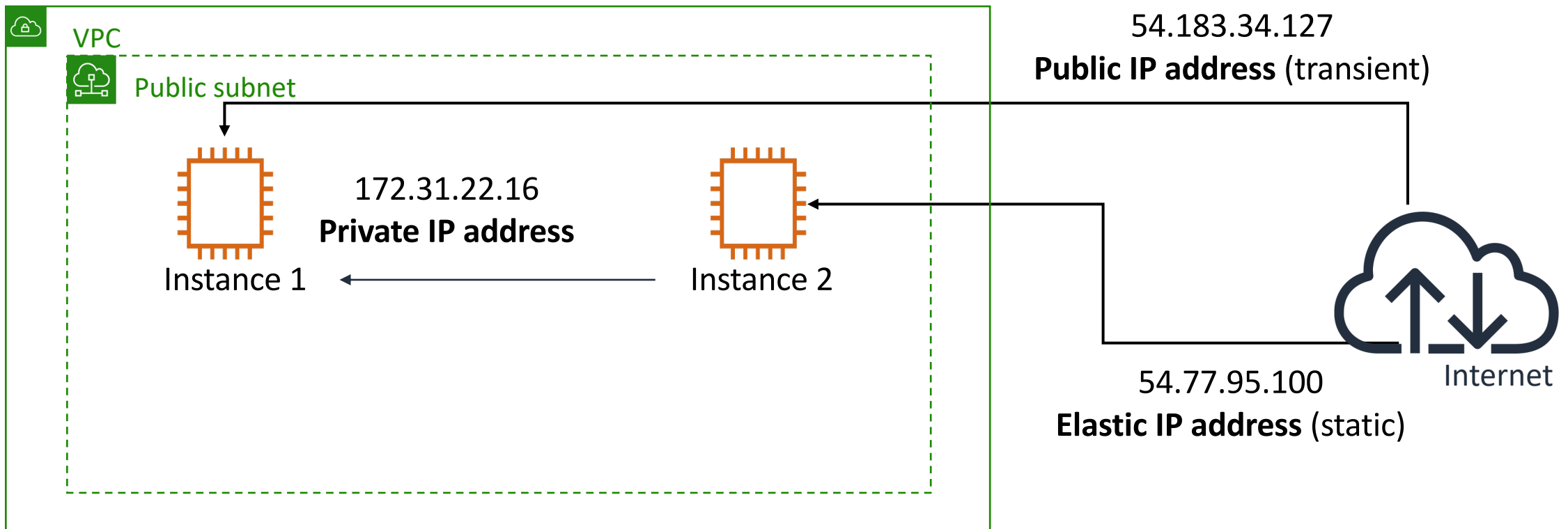
Cold HDD (sc1)
volumes

Amazon EC2 networking



Private, public, and elastic IP addresses

- Public IP addresses change when an instance is stopped and restarted.
- Use Elastic IP addresses for static, public IP addresses.



Security groups

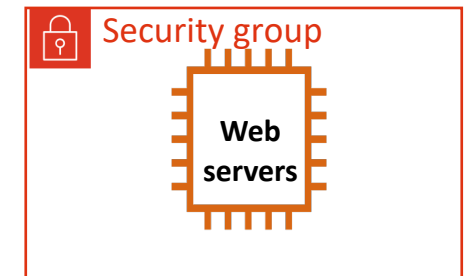
- Restrict access to instances by:
 - Port range
 - IP address range
 - Resource ID
- Instances can be associated with multiple security groups
- Allow data ingress and egress
- Can be added or modified after launch

Example security group settings:

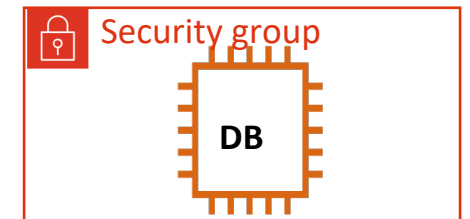
Remote Access on port 22



Remote Access on port 22 and Web traffic 80



Port 3306



Security groups: Examples

Allow HTTP port access **from anywhere**:

ID	Port Range	Source
sg-dfc83cba	80 (HTTP)	0.0.0.0/0

Allow SSH access from a **specific computer**:

ID	Port Range	Source
sg-4ad3712f	22 (SSH)	10.50.2.133/32

Allow SSH access from **members of a security group**:

ID	Port Range	Source
sg-d1cd6fb4	22 (SSH)	sg-4ad3712f

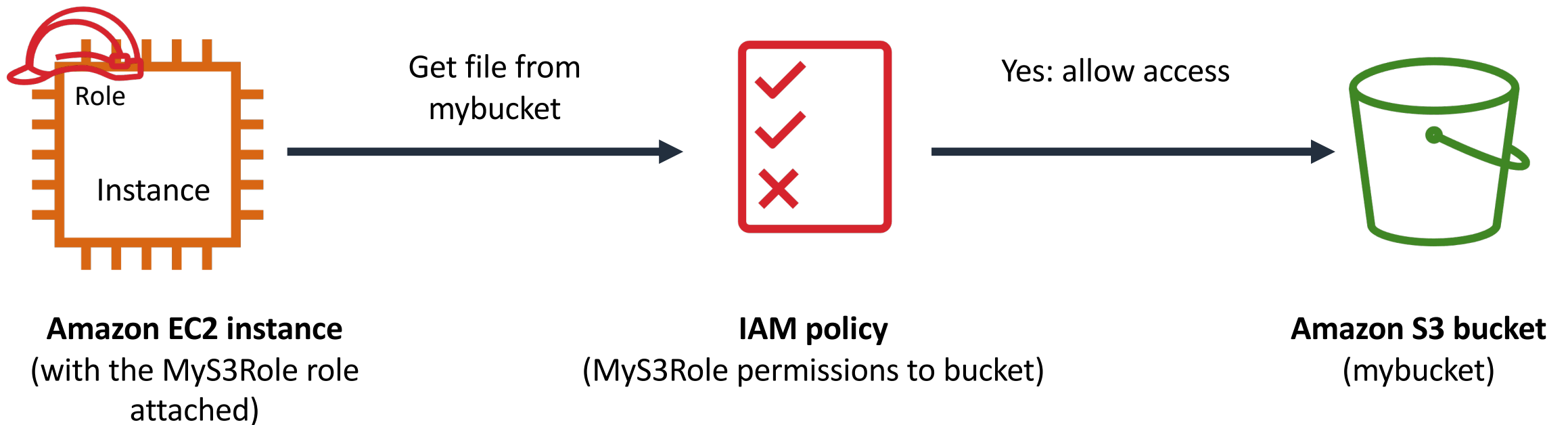
- Avoid storing access key and secret key locally on EC2
- Allows you to attach an IAM role to an EC2 instance
- Automatic propagation of access keys to instances
 - Automatic rotation of access keys multiple times daily
 - Use across multiple instances (e.g., Auto Scaling group)

Example: Grant EC2 instances permission to access the Amazon S3 buckets where application data is stored.

Best Practice

Do not store permanent access keys on EC2 instances, because credentials could be compromised.

Example of instance profile



- Script that runs at launch:
 - Linux script
 - Windows batch or PowerShell scripts
- Script is run by:
 - Cloud-init on Linux
 - EC2Launch service on Windows

Linux Shell Script

```
#!/bin/bash
yum update -y
yum groupinstall -y
    "Web Server" "PHP Support"
yum install -y php-mysql
service httpd start
chkconfig httpd on
```

cloud-init directives

```
#cloud-config
repo_update: true
repo_upgrade: all

packages:
- httpd
- php
- php-mysql

runcmd:
- chkconfig httpd on
```

Batch file

```
<script>
winrm quickconfig -q

winrm set winrm/config/winrs
@{MaxMemoryPerShellMB="300"}

winrm set winrm/config
@{MaxTimeoutms="1800000"}
</script>
```

PowerShell script

```
<powershell>
# Install IIS and Web
# Management Tools.

Import-Module ServerManager

Install-WindowsFeature
    web-server, web-webserver
Install-WindowsFeature
    web-mgmt-tools
</powershell>
```

Amazon EC2 instance metadata

- Data about your instance that you can use to configure or manage the running instance
- Metadata service is provided for every instance at the IP address **169.254.169.254**
- Query instance metadata

```
export INSTANCE_ID=$(curl http://169.254.169.254/latest/meta-data/instance-id)
```

```
aws ec2 describe-instances --instance-id $INSTANCE_ID
```

Linux

```
$INSTANCE_ID=(new-object net.webclient).DownloadString(  
    'http://169.254.169.254/latest/meta-data/instance-id')
```

```
aws ec2 describe-instances --instance-id $INSTANCE_ID
```

Windows

User data and Amazon EC2 instance metadata

- It might be useful to reference properties of the instance in user data.
- Query instance metadata and use in a user data script at launch.

To change hostname of Amazon Linux AMI to instance-id at launch:

User Data Script

Linux

```
#!/bin/bash

#Get hostname from instance-id metadata = $newhost
newhost=$(curl http://169.254.169.254/latest/meta-data/instance-id/)

#change hostname in /etc/hosts & /etc/sysconfig/network
sudo sed -i "s/\<localhost\>/$newhost/g" /etc/hosts
sudo sed -i "s/\<localhost\>/$newhost/g" /etc/sysconfig/network

#Reboot instance
sudo reboot
```

Retrieve EC2 instance user data

Examples of EC2 instance metadata include:

```
[ec2-user ~]$ curl http://169.254.169.254/latest/meta-data/  
ami-id  
ami-launch-index  
ami-manifest-path  
block-device-mapping/  
events/  
hostname  
iam/  
instance-action
```

EC2 instance user data

Example, with result:

```
[ec2-user ~]$ curl http://169.254.169.254/latest/user-data
#!/bin/bash
yum update -y
service httpd start
chkconfig httpd on
```

Additional metadata examples

Example of retrieving the **public-hostname**:

```
[ec2-user ~]$ curl http://169.254.169.254/latest/meta-data/public-hostname  
ec2-203-0-113-25.compute-1.amazonaws.com
```

Example of retrieving the **subnet ID for an instance**:

```
[ec2-user ~]$ curl http://169.254.169.254/latest/meta-data/network/interfaces/macs/02:29:96:8f:6a:2d/subnet-id  
subnet-be9b61d7
```



Remote Access

SSH

Linux

- Use SSH to log into an instance.
- Password-only authentication is disabled on Linux by default.
- Port 22

Windows

RDP

- Use RDP to log into instances.
- Uses an encrypted random password that can only be decrypted by using a private secret access key.
- Port 3389

Setting your own instance security

Change users and permissions after instances are created

- **Windows:**
 - Integrate with AWS Directory Service
- **Linux:**
 - Uses Kerberos-based single sign-on (SSO)
 - Generate a public/private key pair for the user using `ssh-keygen`
 - Add the user's public key to their `~/.ssh/authorized_keys`

Devise a strategy for fleet-wide user management:

- Creating AMIs
- Using configuration software

Configuration management applications can:

- Simplify the process of granting and revoking access across a large number of machines
- Reduce the process down to a few simple commands

Configuration management applications:

- Chef
- Puppet
- Ansible

Post-launch configuration options

Get instance screenshot

**Enable termination
protection**

**Disable
source/destination check**
(enabled by default) if the
instance is performing a
NAT function as a router or
firewall

Launch Amazon EC2 in the AWS CLI

```
aws ec2 run-instances
--image-id ami-0123456789012345
--instance-type t2.micro
--key-name mykeypair
--security-group-ids sg-0123456789012345
--subnet-id subnet-0123456789012345
--iam-instance-profile Name=EC2Admin
--user-data file://UserData.txt
--tag-specifications 'ResourceType=instance,Tags=[{Key=Name,Value=WebServer}] '
```

Demo: Launch Amazon EC2 instance

Key Takeaway

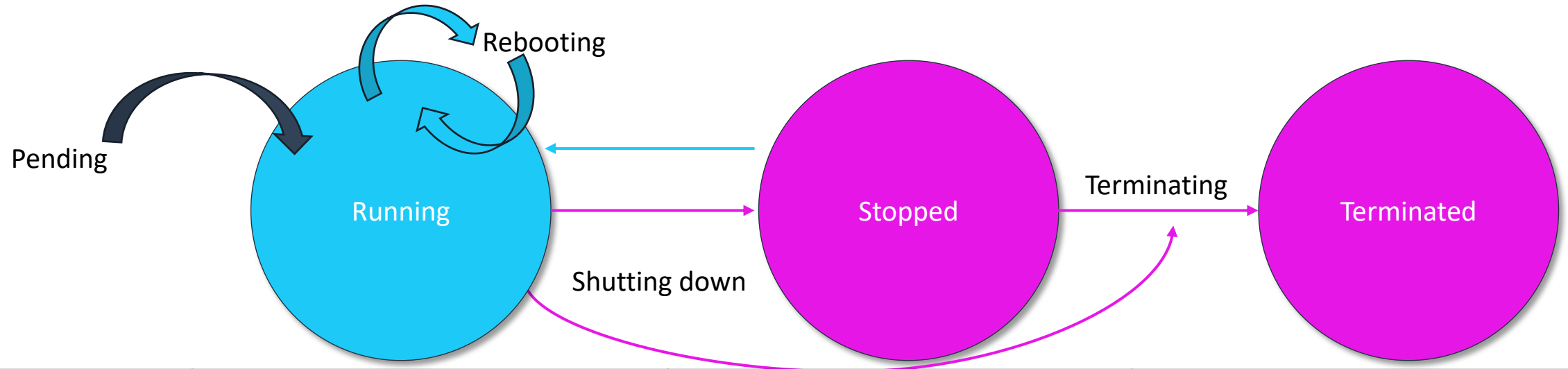


- Do not store permanent access keys on EC2 instance profile
- Choose an instance type that best suits your application requirement. If required, change instance type in future.
- Elastic Block Storage (EBS) options include:
 - Solid state drives (SSD)
 - Hard disk drives (HDD)
- EC2 instance metadata can be accessed programmatically to configure or manage the instance.

Module 3: Computing (Servers)

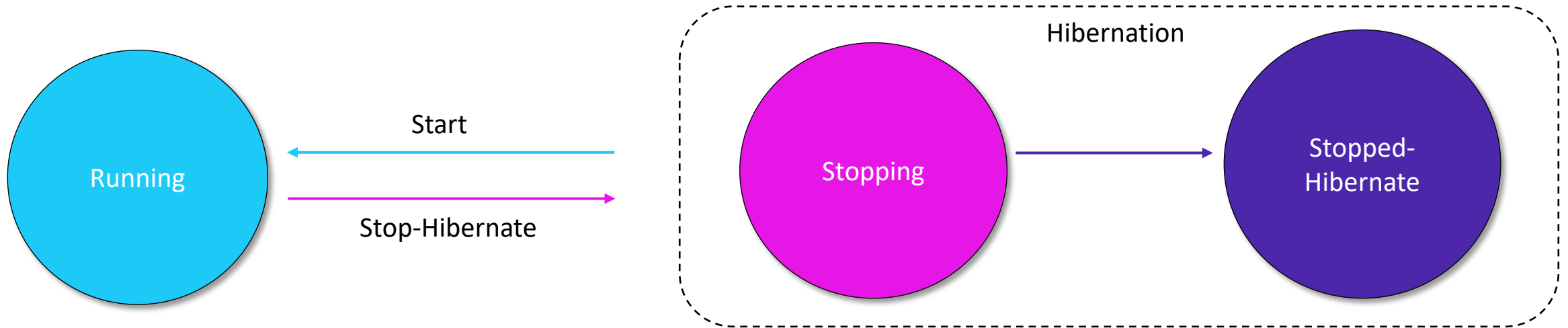
Managing your AWS instances

Lifecycle states of an EC2 instance



State	Running	Stopped	Terminated
Charged?	Charged	Not charged	Not charged
Instance Store	Retained	Lost	Lost
EBS Volumes	Retained	Retained	Deleted if DeleteOnTermination is set
Public IP Address	Retained	Released	Released
Elastic IP Address	Retained	Retained	Disassociated

Hibernation of an EC2 instance



	Running	Stop-Hibernate
Charged?	Charged	Not charged*
Instance Store	You cannot stop or hibernate instance store-backed instances.	
EBS Volumes	Retained	Retained
Public IP Address	Retained	Released
Elastic IP Address	Retained	Retained
Private IP	Retained	Retained

Hibernation: Prerequisites

To enable
hibernation
at launch,
you require:

A supported instance type and AMI

An encrypted root EBS volume

On-Demand, Spot, or Reserved Instances

NOTE: Hibernation cannot be enabled on an existing Amazon EC2 instance after launch.

Common reasons why instances might need to be relaunched:

Auto Scaling

Cost savings

Downgrading

**Repairing
impaired
instances**

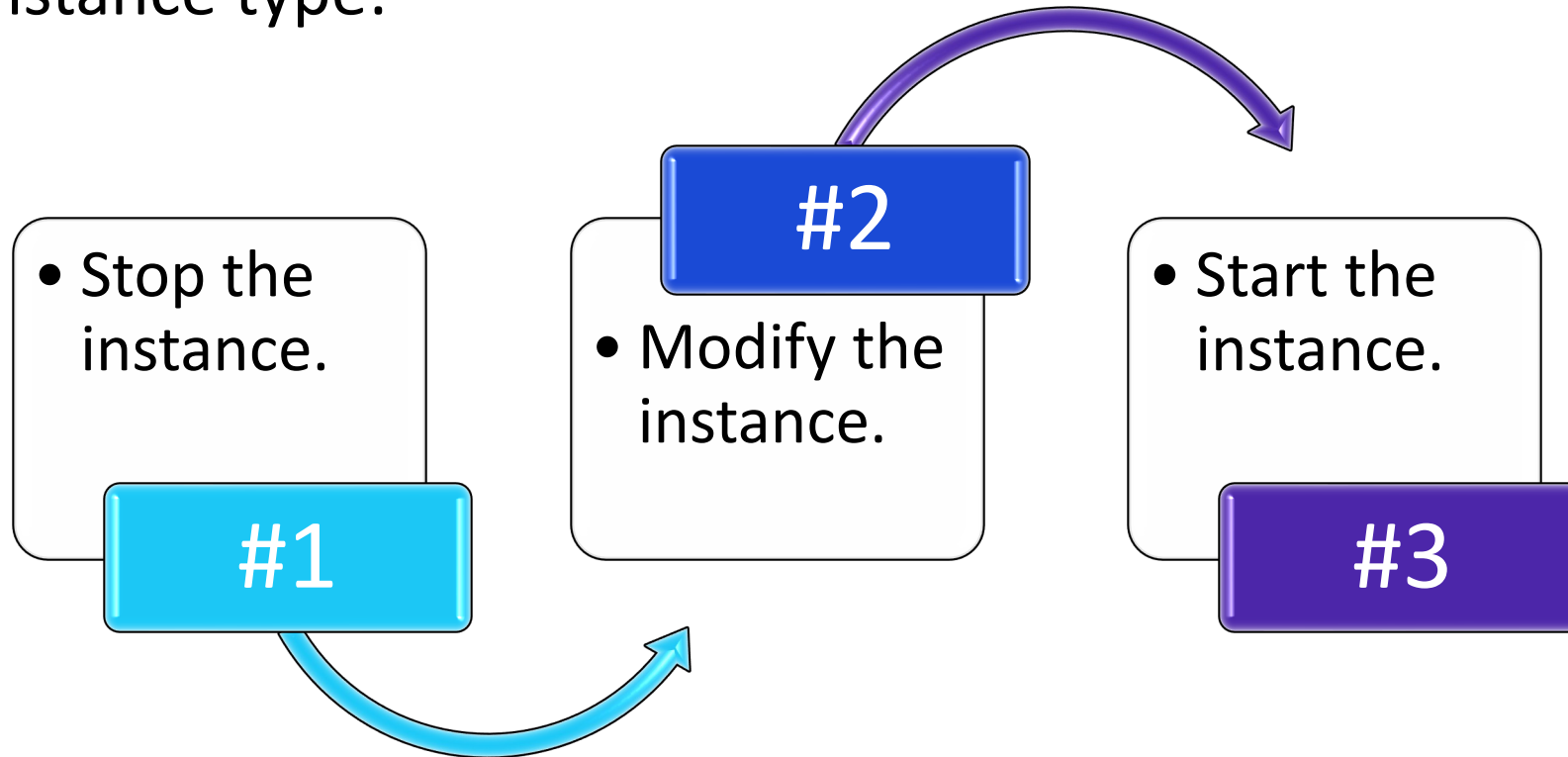
**Upgrading OS
architecture or
image type**

Best Practice

Design your instances for easy build-up and tear-down.

Transitioning to a new instance size

To modify instance type:



```
aws ec2 modify-instance-attribute --instance-type m5.xlarge
```

AMI deprecation does not affect running instances:

- **Windows AMIs**
 - Deprecated soon after release of new image
 - AMIs released within 5 days of Microsoft's patch
 - Find by name, not AMI ID
- **Amazon Linux AMIs**
 - Available for years

Query the latest Amazon Linux AMI

```
aws ssm get-parameters-by-path --path "/aws/service/ami-amazon-linux-latest"  
--query "Parameters[?Name=='/aws/service/ami-amazon-linux-latest/amzn-ami-  
hvm-x86_64-gp2'].Value" --output text
```

Updating running EC2 instances

- **Windows:**

- The instance can use Windows Update
- Or, update manually

- **Amazon Linux:**

- The instance will install security updates on initial launch.

Additional tools for managing Amazon EC2 instance updates:

- AWS Systems Manager or AWS OpsWorks
- Third-party configuration management tools



AWS Systems Manager



AWS OpsWorks

Key Takeaway



- Understand the different states of an EC2 instance:
 - Running
 - Stopped
 - Terminated
 - Hibernation
- Know the implication of instance state on billing.
- If the hibernation option is needed, it must be configured at launch.

Module 3: Computing (Servers)

Securing your AWS instances



The AWS shared responsibility model

C
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Customer applications and content

Platform, applications, identity and access management

Amazon EC2 operating system, network, and firewall configuration

Client-side data encryption

Server-side data encryption

Network traffic protection

AWS Foundation Services

Compute

Storage

Database

Networking

AWS global infrastructure

Availability Zones

Regions

Edge locations



Module 3: Computing (Servers)

Amazon EC2 purchasing options

Amazon EC2 purchasing options (1 of 2)

Instance Type	Benefit	Use Case
On-Demand	• Low cost and flexibility	• Short term, spiky, or unpredictable workloads • Applications being developed or tested
Reserved	• No Upfront RI does not require an upfront payment • Up to 75% discount compared to On-Demand pricing and capacity reservations	• Steady state or predictable usage • Require reserved capacity
Spot	• Flexible start and end times • Feasible at very low compute prices	• Urgent computing needs with steep discounts • Large amounts of additional capacity

Amazon EC2 purchasing options (2 of 2)

Instance Type	Benefit	Use Case
Dedicated Hosts	- Physical host fully dedicated to running your instances - You can bring your per-socket or per-VM software licenses to reduce cost	You want to use your existing per-socket, per-core, or per-VM software licenses
Dedicated Instances	Instances run on hardware that is dedicated to a single account	When regulatory restrictions that require physical isolation apply
i3.metal	Physical host on which you can run the operating system or virtualization layer of your choice	Workloads that require access to hardware feature sets (such as Intel® VT-x)

- Specify interruption behavior:
 - Hibernate: saves memory state to disk (up to 122GB)
 - Stop
 - Terminate
- Interruption warning: Instance receives 2 minute notice.
- Hook operating system shutdown events to clean up:
 - **Linux:** Hook into runlevels (/etc/rc0.d, /etc/rc6.d)
 - **Windows:** Add script to Shutdown event in local computer policy

Best Practice

**Use encrypted root Amazon EBS volumes
to secure state information.**

- Amazon EC2 virtualization
- Instance types and storage options available for EC2 instances
- Networking components that must be specified when launching an EC2 instance
- Amazon EC2 user data and metadata
- The lifecycle states for an EC2 instance
- The shared responsibility model
- How to create Amazon EC2 instances

To finish this module:

Complete:



Knowledge check

Test Strategies



Module 3 scenario:

A SysOps engineer working at a local university has decided to launch Amazon EC2 instances to reduce their hardware footprint. They are new to AWS and will only need to launch in one region until their business grows.

What is the first step when you launch an Amazon Elastic Compute Cloud (Amazon EC2) instance?



Module 3 scenario:

A SysOps engineer working at a local university has decided to launch Amazon EC2 instances to reduce their hardware footprint. They are new to AWS and will only need to launch in one region until their business grows.

What is the first step when you launch an Amazon Elastic Compute Cloud (Amazon EC2) instance?

Module 3 scenario:

A SysOps engineer working at a local university has decided to launch Amazon EC2 instances to reduce their hardware footprint. They are new to AWS and will only need to launch in one region until their business grows.

What is the first step when you launch an Amazon Elastic Compute Cloud (Amazon EC2) instance?

- A. Determine the Amazon Machine Image (AMI) that will be used.
- B. Launch access and control permissions.
- C. Provide configuration details.
- D. Select instance type and size.

Lab 2: Creating Amazon EC2 instances



Lab 2: Creating Amazon EC2 instances



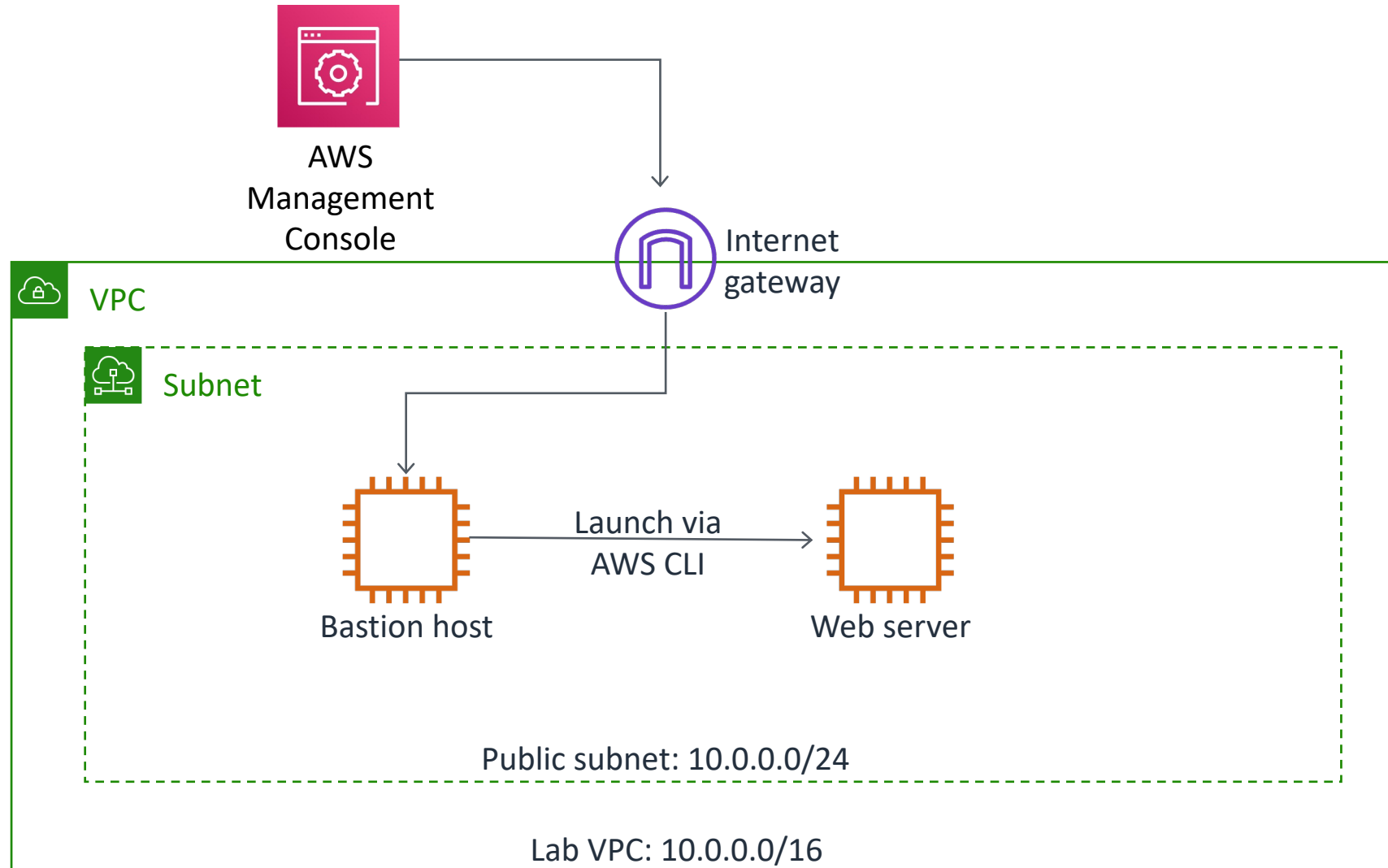
In this lab, you will:

- Launch an Amazon EC2 instance using the AWS Management Console.
- Launch an Amazon EC2 instance using the AWS Command Line Interface (AWS CLI).

Lab 2: Amazon EC2

1. Launch an Amazon EC2 instance by using the **AWS Management Console**.
2. Launch an Amazon EC2 instance by using the **AWS Command Line Interface (AWS CLI)**.
3. Can also complete two additional **challenges** to diagnose and fix problems

Lab 2: Creating Amazon EC2 instances

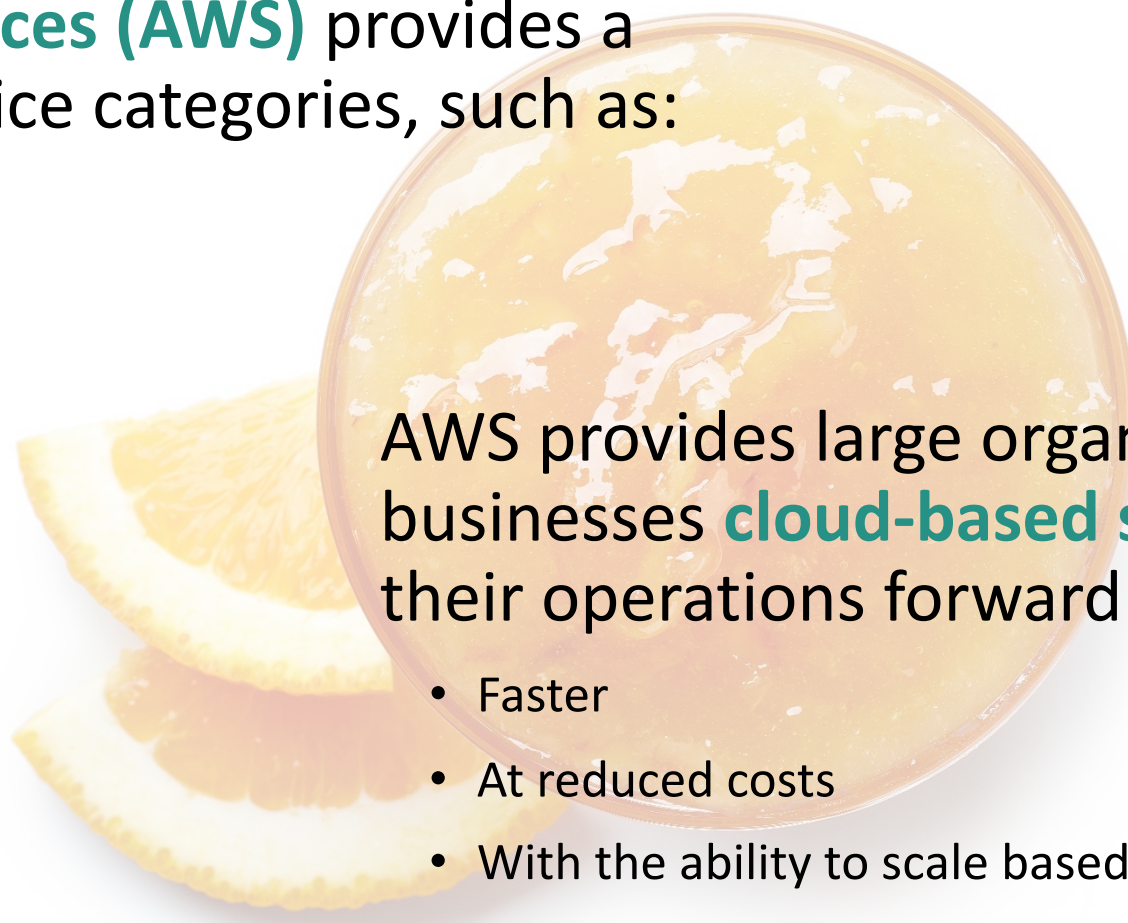


Systems Operations Project Scenario

Mom & Pop Café & Bakery

Amazon Web Services (AWS) provides a broad array of service categories, such as:

- Compute
- Storage
- Database
- Analytics

A photograph of a glass bowl filled with orange juice, with several orange slices floating in it. The bowl is placed on a white surface, and the background is a soft, out-of-focus white.

AWS provides large organizations and small businesses **cloud-based solutions** to move their operations forward:

- Faster
- At reduced costs
- With the ability to scale based on demand

On completion of this project, you will be able to:

- **Experience** the communication challenges faced when attempting to apply technology as the solution to business problems.
- **Translate** customer requirements into a proposed technical solution.
- **Present** the proposed solution to the customer.
- **Use** AWS Best Practices for a frugal, resilient, and secure solution.



Mom & Pop Café & Bakery

Customer Meeting Preparation

Mom & Pop Café & Bakery

Request for the ability to place online orders

Business need: Ability to order online



Darsh thought back to when Matt stopped by the other day for his coffee. Darsh knows that Matt is always running between meetings, and thought it would be helpful if customers could order online.

Darsh mentions his idea to Phillip, Emma, and Sophie to get their thoughts.

Online ordering is a requested feature



While in the café, customers have expressed:

- Appreciation for the website and the information it provides
- Interest in being able to place orders online

Online ordering is live

Sophie and Darsh redeployed the website so that it:

- Has a database backend
- Maintains order history

However, there were some issues when trying to deploy it.



Matt offers to help

Matt comes back at lunch to walk Sophie and Darsh through some troubleshooting procedures so they can find the issues.

Matt explains that what they created was a **Linux, Apache, MariaDB, PHP (LAMP) stack**:

- Based on a set of open source software
- Installed on a Linux server
- Used to create a website that has a database backend

Learn more about
the LAMP stack

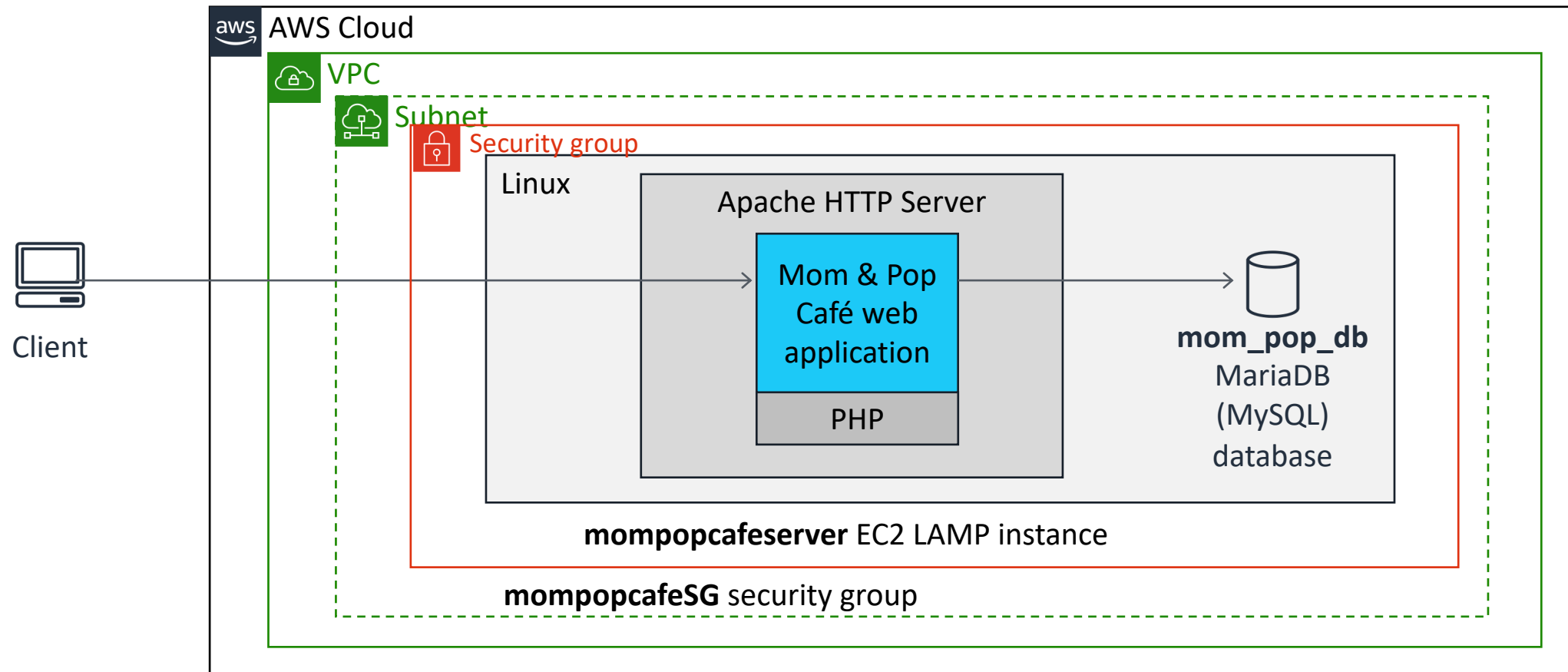


Activity 3: Troubleshoot creating a LAMP instance via the AWS CLI



Activity 3: Troubleshoot creating a LAMP instance via the AWS CLI

This activity will require some troubleshooting to successfully deploy the version of the Mom & Pop Café website that supports online orders.





Begin Activity #3: Deploy a LAMP Instance

Database backend features

What feature enhancements might be gained by the fact that the website now has a database backend?

Enhancement #1

Enhancement #2

Enhancement #3

Database backend features

What feature enhancements might be gained by the fact that the website now has a database backend?

Enhancement #1

Customer orders can be placed and tracked.

Enhancement #2

The website can prompt for customer names and contact info, which could help marketing.

Enhancement #3

The business will have a record of sales, and could glean insight from the data.

Project time: Troubleshooting knowledge base



Thank you

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